Examen Ordinario
Periodo 202610

Materia: Data Mining
Semestre: January - May 2026
Profesor: Víctor Cardoso Fernández
Calificación: ____________

ALUMNO: ______________________________ **Fecha:** May 14, 2026

Description

The objective of this final project is to apply the predictive modeling techniques learned throughout the course to a real dataset developed by your team. Each team will use the same dataset they have been working on during the semester and apply multiple models to analyze and solve a specific problem.

You are expected to:

- Define a clear problem based on your dataset
- Select appropriate input variables (X) and a target variable (y)
- Apply multiple predictive models covered in the course
- Evaluate and compare model performance
- Interpret the results and identify the best model for your problem

Models to Apply

Your project must include the implementation of multiple predictive models using the same dataset and target variable, so that results can be compared consistently.
Follow these guidelines carefully:

1. Decision Tree Model

You must:

- Use a Decision Tree model (e.g., DecisionTreeRegressor or DecisionTreeClassifier, depending on your target variable)
- Train the model using your dataset
- Apply at least one complexity control parameter, such as:
    - max_depth
    - min_samples_split

You are expected to:

- Train at least two versions of the model:
    - One without constraints (default parameters)
    - One with controlled complexity
- Compare both versions and briefly explain:
    - How the parameter affected the model
    - Whether performance improved or not

2. Regression Models

You must implement both:

a) Linear Regression

- Use a standard linear regression model

- Train the model using your dataset
- Evaluate its performance using the required metrics

You should:

- Clearly define your input variables (X)
- Clearly define your target variable (y)

b) Polynomial Regression

- Transform your input variables using polynomial features (e.g., degree = 2 or 3)
- Train a regression model using these transformed variables

You are expected to:

- Compare results between:
    - Linear regression
    - Polynomial regression
- Explain:
    - Whether the polynomial model improves performance
    - Why this improvement happens (or not)

3. Artificial Neural Network (ANN)

You must:

- Implement an ANN model (e.g., MLPRegressor)
- Scale your input variables before training the model

You are required to:

- Train at least two different ANN configurations, modifying parameters such as:
    - hidden_layer_sizes
    - activation
    - max_iter

You should:

- Compare the performance of both configurations
- Explain:
    - How changing parameters affects the results
    - Which configuration performs better and why

Important Guidelines for All Models

For every model you implement:

- Use the same dataset and target variable
- Use a train/test split (e.g., 80/20)
- Ensure that all models are evaluated under the same conditions

This is essential for a fair comparison.

Final Comparison Requirement

After implementing all models, you must:

- Compare their performance using:
  - R^2
  - MAE
  - MSE
  - RMSE
- Support your analysis with:
  - Predicted vs Actual plots
  - Residual plots
- Clearly answer:

Which model performs best for your dataset, and why?

Key Expectation

You are not only expected to run the models, but to demonstrate understanding of how and why their performance differs.

Deliverables

1. GitHub Repository (40%)

Each team must upload to their assigned GitHub folder:

- The dataset used in the project
- All Python scripts implementing the models
- Clean and well-organized code
- Clear comments explaining key steps

The code must:

- Run correctly
- Include all required models
- Be structured in a clear and readable way

2. Written Report (30%)

The report must include the following sections:

- **Introduction**
  - Problem description
  - Objective of the project
- **Methodology**
  - Description of each model applied
  - Differences between models

  - Explanation of selected variables
- **Results**
  - Performance metrics for each model
  - Graphical analysis
  - Model comparison
- **Conclusions**
  - Best model identified
  - Explanation of results
  - Limitations and possible improvements

3. Presentation (30%)

- Maximum duration: **10 minutes per team**
- All team members must participate

The presentation must include:

- Problem description
- Dataset overview
- Models applied
- Results comparison
- Final conclusions

Final Note

All deliverables must be consistent with each other.

- The models shown in the presentation must match the code
- The results in the report must match the outputs from the scripts
- The dataset must be the same across all deliverables

.

Examen Ordinario
Periodo 202610

Materia: Data Mining
Semestre: January - May 2026
Profesor: Víctor Cardoso Fernández **Calificación:** ____________

ALUMNO: ______________________________ **Fecha:** May 14, 2026

Evaluation rubric: GitHub Repository (40%)

Criteria	Excellent (3)	Good (2)	Fair (1)	Needs improvement (0)
Implementation of Required Models	All required models are correctly implemented and extended with the required variations (e.g., tree pruning comparison, polynomial degree comparison, ANN hyperparameter tuning). The implementation demonstrates a strong understanding of each model and its configuration.	All required models are implemented correctly, including Decision Tree, Linear Regression, Polynomial Regression, and ANN. Basic functionality is achieved, and models run as expected, but without deeper exploration or variation.	Some models are implemented, but key components are missing or incorrectly applied. For example, missing polynomial transformation, no ANN variation, or incorrect use of decision tree parameters. Models may run but do not follow the instructions given in the project.	The repository does not include the required models, or the implementations are incorrect to the point that they do not reflect the expected methodologies (Decision Tree, Regression, ANN). Code may be incomplete, copied without adaptation, or non-functional.
Correct Use of Dataset	The dataset is used consistently and is well-prepared (e.g., clear structure, appropriate variable selection, handling of categorical variables if needed). The target variable is clearly defined and justified, ensuring	The same dataset and target variable are used consistently across all models. Data is correctly loaded and used in a standard format, but without additional preparation or validation.	The dataset is present but inconsistently used across models (e.g., different preprocessing, different variables, or incorrect splits). The target variable may be ambiguously defined.	The dataset is missing, incorrectly uploaded, or not used consistently across models. The target variable is unclear or incorrectly defined.

Examen Ordinario
Periodo 202610

Materia: Data Mining
Semestre: January - May 2026
Profesor: Víctor Cardoso Fernández
Calificación: ____________

ALUMNO: ______________________________
Fecha: May 14, 2026

	a fair and meaningful comparison across models.			
Model Evaluation (Metrics)	Metrics are correctly calculated and clearly structured (e.g., tables or organized output) to facilitate direct comparison across all models. The implementation shows a clear intention to support analysis and decision-making.	All required metrics (R^2, MAE, MSE, RMSE) are correctly calculated for each model. Results are valid and allow for a basic comparison between models.	Some metrics are calculated, but there are inconsistencies or errors (e.g., missing metrics, incorrect formulas, or metrics not applied to all models). Comparison is incomplete.	Evaluation metrics are missing, incorrectly calculated, or not aligned with the models. Results are not interpretable or are clearly incorrect.
Graphical Analysis	Graphical analysis is correctly implemented for all models. Plots are clear, well-labeled, and useful for comparing model performance and identifying patterns such as bias, dispersion, or errors.	Required plots (Predicted vs Actual and Residual plots) are correctly generated for at least one model and are readable and properly labeled.	Graphical outputs are present but incomplete, poorly labeled, or difficult to interpret. Required plots may be missing or incorrectly implemented.	No graphical analysis is included, or the plots are incorrect or not related to model evaluation.
Reproducibility	The code runs cleanly from start to finish, with all required files included. Results are consistent and reproducible, demonstrating a complete and reliable implementation.	The code runs correctly using the provided dataset and produces valid results without major issues.	The code runs partially but requires fixes, manual adjustments, or missing files to execute correctly. Results are inconsistent or unclear.	The code does not run, or critical components (dataset, scripts) are missing, making reproduction impossible.

Examen Ordinario
Periodo 202610

Materia: Data Mining
Semestre: January - May 2026
Profesor: Víctor Cardoso Fernández **Calificación:** ____________

ALUMNO: ______________________________ **Fecha:** May 14, 2026

Evaluation rubric: Written Report (30%)

Criteria	Excellent (3)	Good (2)	Fair (1)	Needs improvement (0)
Introduction (Problem Definition and Objective)	The problem is clearly defined and well contextualized. The objective is specific, well justified, and clearly connected to the dataset and the purpose of the analysis.	The problem is clearly defined, and the objective of the project is stated. The context is understandable, although justification or depth may be limited.	The problem is mentioned but poorly defined. The objective is vague, incomplete, or not clearly connected to the dataset. The context lacks clarity or relevance.	The introduction is missing or does not clearly define the problem. There is no clear objective, and the context of the project is not understandable.
Methodology (Models and Approach)	The methodology is clearly structured and detailed. It explains each model, highlights the differences between them, and justifies key decisions such as variable selection, model configuration, and preprocessing steps.	The methodology clearly describes the models used (Decision Tree, Regression, ANN) and the general approach. Steps are understandable, but explanations may lack depth or comparison between models.	The methodology is partially described, but key elements are missing (e.g., unclear explanation of models, missing preprocessing steps, or lack of structure).	The methodology section is missing or does not describe the models used. There is no explanation of how the analysis was conducted.
Results and Presentation of Findings	Results are well organized and clearly presented (e.g., tables, structured outputs). Metrics are complete and	Results are clearly presented for each model, including the required metrics. The	Results are partially presented, but there are inconsistencies, missing metrics, or unclear	Results are missing or incorrectly presented. Metrics and outputs are

	consistently reported across models, making comparison straightforward and meaningful.	structure allows for basic comparison between models.	organization. Comparison between models is difficult.	not included or are not understandable.
Analysis and Model Comparison	Strong and well-reasoned analysis. Models are compared using metrics and graphical insights. The best model is clearly identified and justified with logical and data-driven arguments.	Clear comparison between models based on metrics and basic interpretation. The best model is identified, but justification may be limited.	Limited or superficial analysis. The comparison between models is weak, unclear, or not supported by evidence.	There is no analysis or comparison between models. The report only presents results without interpretation.
Conclusions and Overall Quality	Conclusions are insightful and clearly supported by the analysis. The report reflects critical thinking, includes limitations and possible improvements, and is well written, structured, and professional.	Conclusions are clear and based on the results obtained. The report is understandable and reasonably well structured.	Conclusions are present but weak, generic, or not clearly supported by the results. Writing may lack clarity or structure.	Conclusions are missing or unrelated to the results. The report is poorly written and difficult to understand.

UNIVERSIDAD ANAHUAC MAYAB

División de
Ingeniería y
Ciencias Exactas

Examen Ordinario
Periodo 202610

Materia: Data Mining
Semestre: January - May 2026
Profesor: Víctor Cardoso Fernández
Calificación: ____________

ALUMNO: ______________________________
Fecha: May 14, 2026

Evaluation rubric: Presentation (30%)

Criteria	Excellent (3)	Good (2)	Fair (1)	Needs improvement (0)
Problem Explanation and Context	The problem is clearly explained and well contextualized. The objective is specific, meaningful, and clearly connected to the dataset and the analysis performed.	The problem and objective are clearly explained. The audience can understand the purpose of the project, although the explanation may lack depth or context.	The problem is mentioned but poorly explained. The context is vague, and the objective of the project is not clearly communicated.	The problem is not explained or is completely unclear. The audience cannot understand what the project is about or why it is relevant.
Explanation of Models and Methodology	The models and methodology are explained clearly and confidently. Differences between models are highlighted, and the audience understands how and why each model was applied.	All models are explained clearly (Decision Tree, Regression, ANN). The general methodology is understandable, although explanations may be basic.	Some models are mentioned, but the explanation is incomplete or confusing. Key aspects of the methodology are missing.	The models are not explained, or the explanation is incorrect. It is unclear what was done or how the models were applied.
Results Interpretation and Model Comparison	Results are clearly explained with strong interpretation. Models are compared using metrics and visual insights, and the	Results are clearly presented and interpreted. Models are compared using metrics, and the best model is identified.	Results are presented but poorly explained. The comparison between models is unclear or not supported by evidence.	Results are not presented or not interpreted. There is no comparison between models.

Red de Universidades Anáhuac

	best model is justified with clear, logical reasoning.			
Visual Quality and Communication	Slides are professional, visually appealing, and well structured. Visual elements (graphs, comparisons) are effectively used to support the message. Communication is clear, concise, and engaging.	Slides are clear, organized, and readable. Visuals (tables, graphs) support the explanation. Communication is understandable.	Slides are basic and contain excessive text or unclear visuals. Communication lacks clarity or structure.	Slides are poorly designed, unreadable, or disorganized. Presentation is difficult to follow.
Team Participation and Time Management	All team members participate actively and evenly. The presentation is well-paced, structured, and fits within the time limit effectively.	Most team members participate. Presentation stays within the 10-minute limit.	Limited participation (one or two members dominate). Time is not well managed.	Only one team member presents, or the presentation is significantly over/under time.